# Full Stack Development with MERN

# Project Documentation format

## 1. Introduction

- **Project Title:shopEZ-MERN E-commerce Application**
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  **2.**Sowmya Kodali
  **3.**Sai Sreedevi Pulaparthi
  **4.** Teomayee Hema Sridevi Geddada

## 2. Project Overview

- **Purpose:** ShopEZ is a full-stack MERN-based e-commerce web application designed to provide customers with a seamless online shopping experience. The platform enables users to browse products, search and filter items, manage shopping carts and wishlists, place orders, and track purchases. It also provides administrative functionalities for managing products, categories, users, and customer orders.

**Features:**

User Registration and Login

JWT-based Authentication

Product Management

Category Management

Product Search

Product Filtering

Shopping Cart

Wishlist

Buy Now

Order Management

Admin Dashboard

## 3. Architecture

• **Frontend:**The frontend is developed using React.js following a component-based architecture. Routing is handled using React Router DOM, while Axios is used for API communication. CSS Modules provide modular styling, and Context API manages authentication, cart, and wishlist states.

• **Backend:**

The backend follows the REST API architecture using Node.js and Express.js. Business logic is separated into controllers and services, while middleware handles authentication, authorization, and error handling.

**Database:**

The application uses MongoDB Community Server, a NoSQL database, to store application data. Mongoose is used as the Object Data Modeling (ODM) library for defining schemas and interacting with the database.

## 4. Setup Instructions

• **Prerequisites:**

**Before running the application, install:**

- Node.js
- npm
- MongoDB Community Server
- Git
- Visual Studio Code

**Installation:**

1.Clone the repository

2.Navigate to the backend folder

3.Install backend dependencies

4.Navigate to the frontend folder

5.Install frontend dependencies

6.Create the backend `.env` file

7.Create frontend `.env`

## 5. Folder Structure

• **Client:**

**client**

**│**

**├── api**

**├── assets**

**├── components**

**├── context**

**├── hooks**

**├── pages**

**├── routes**

**├── services**

**├── App.jsx**

**└── main.jsx**

**• Server:**

server

├── config
├── controllers
├── middleware
├── models
├── routes
├── services
├── utils
├── uploads
├── app.js
└── server.js

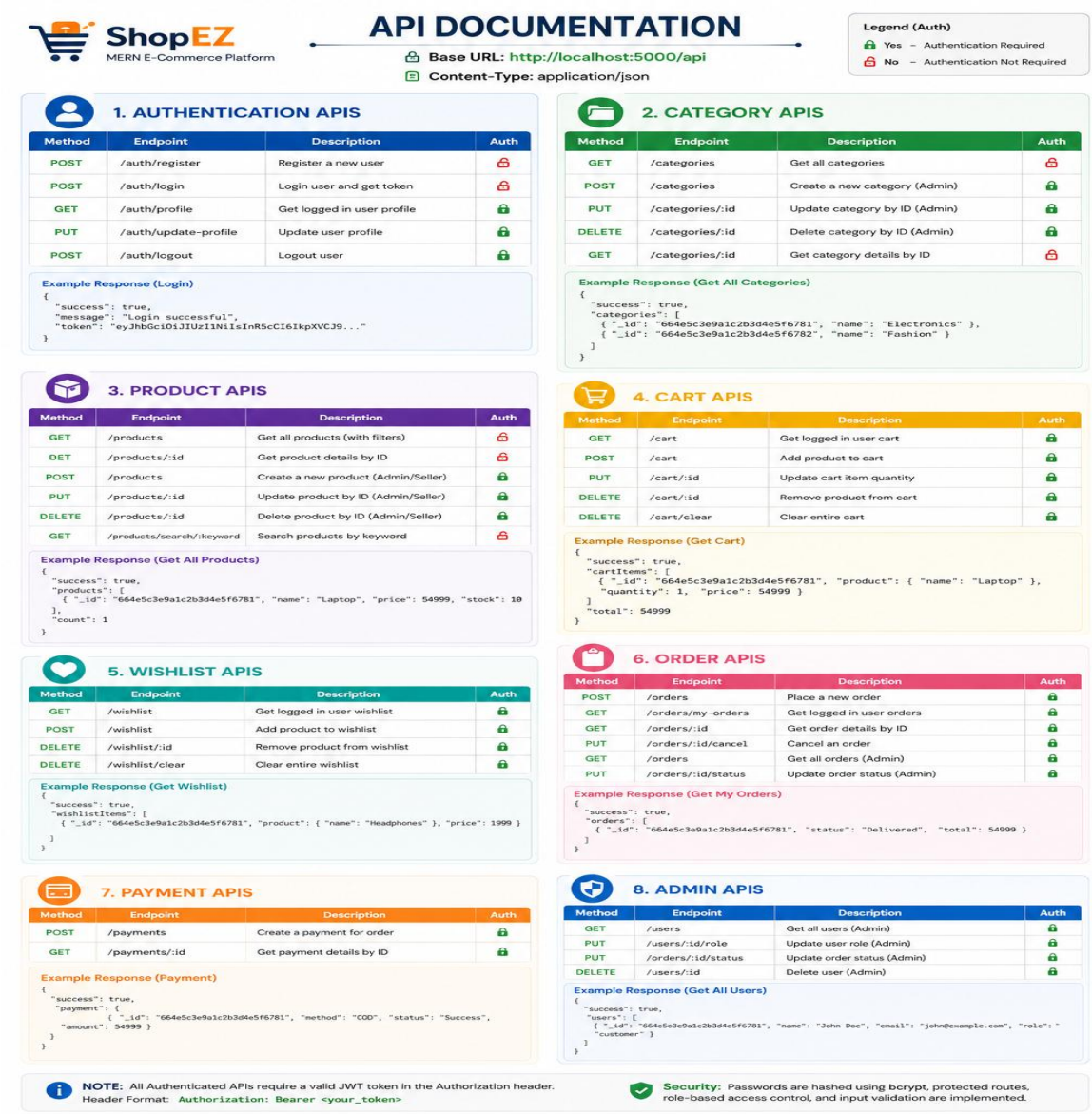
## 6. Running the Application

• Provide commands to start the frontend and backend servers locally. ○

**Frontend: cd client,**`npm run dev` in the client directory.

○ **Backend:** `cd server,npm start in server directory`

## 7. API Documentation




## 8. Authentication

### Authentication Flow:

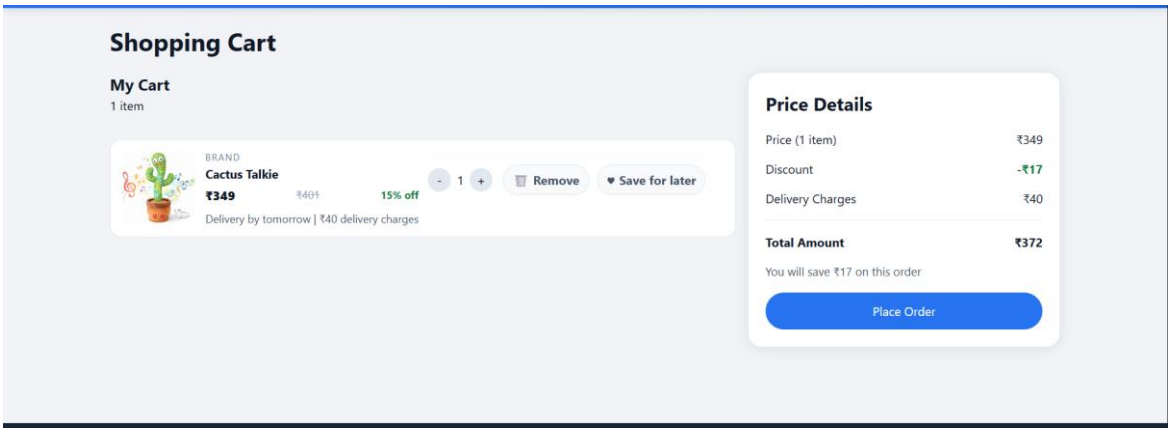
1. User registers an account.

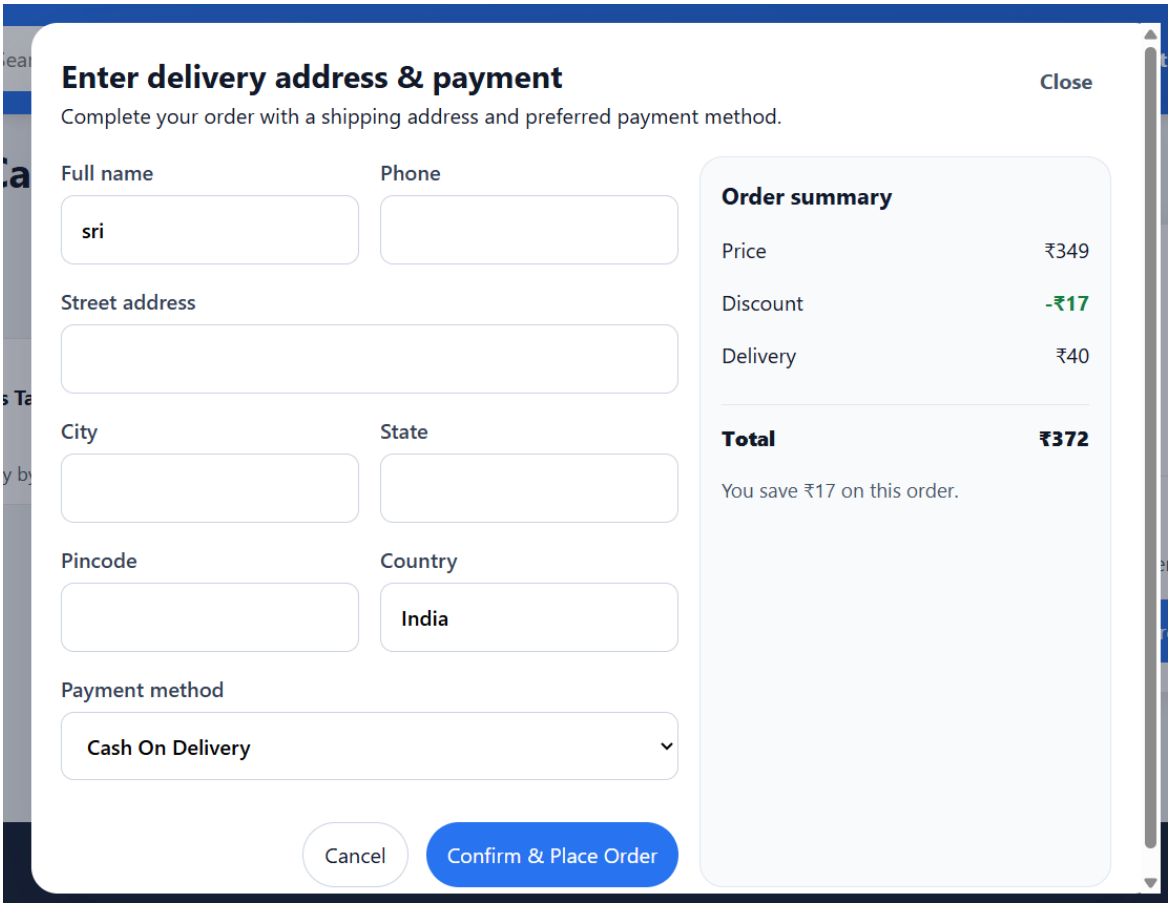
2. User logs in using email and password.
3. Backend verifies credentials.
4. JWT token is generated.
5. Token is stored in Local Storage.
6. Axios automatically sends the token in the Authorization header.
7. Protected routes verify the token before granting access.
8. Role-based authorization restricts access to Admin and Seller features.

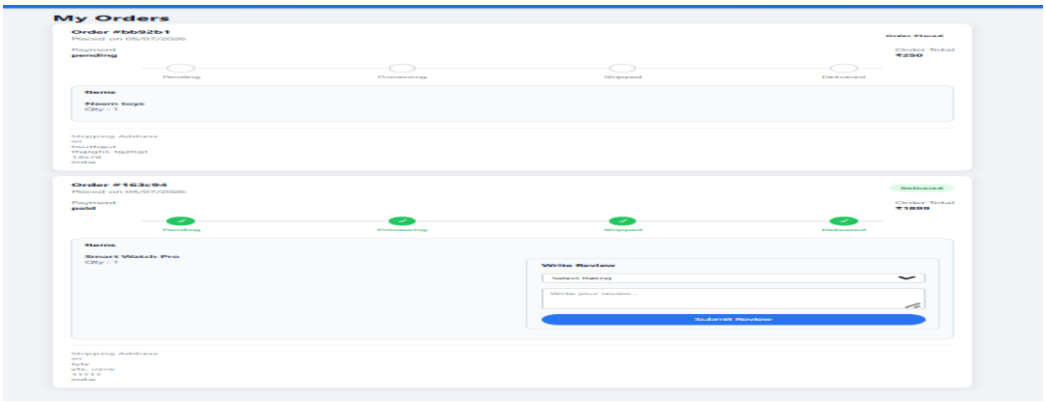
**Security measures include:**

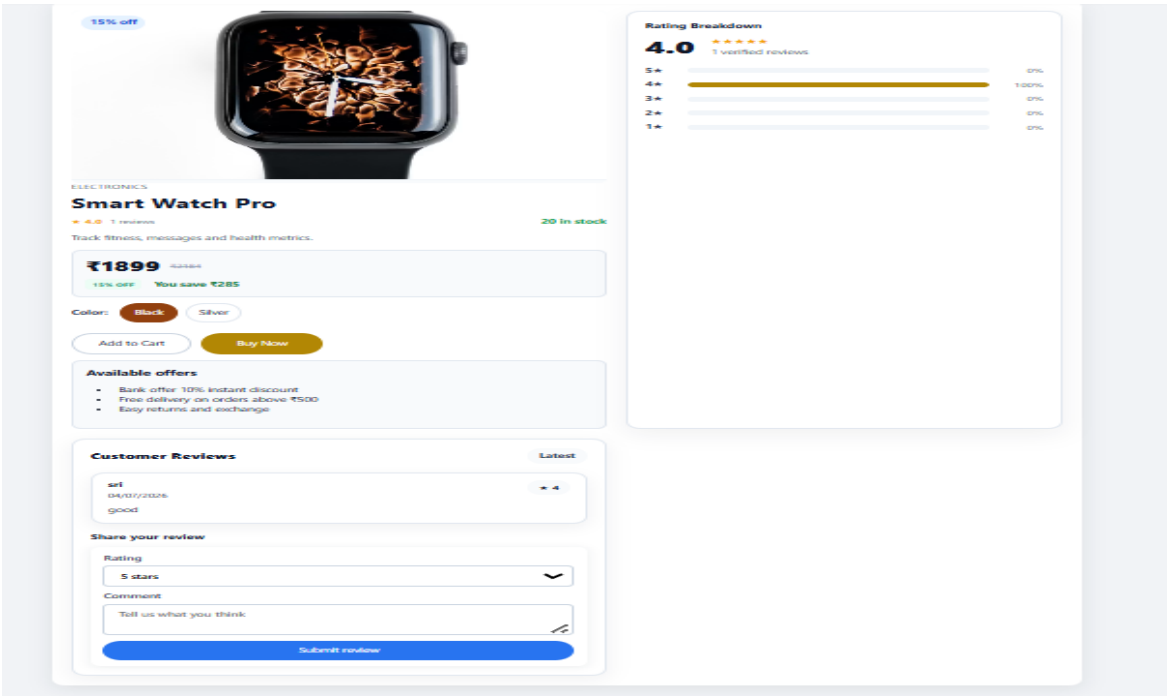
- Password hashing using bcrypt
- JWT Authentication
- Protected REST APIs
- Role-Based Access Control
- Environment Variables
- CORS Protection

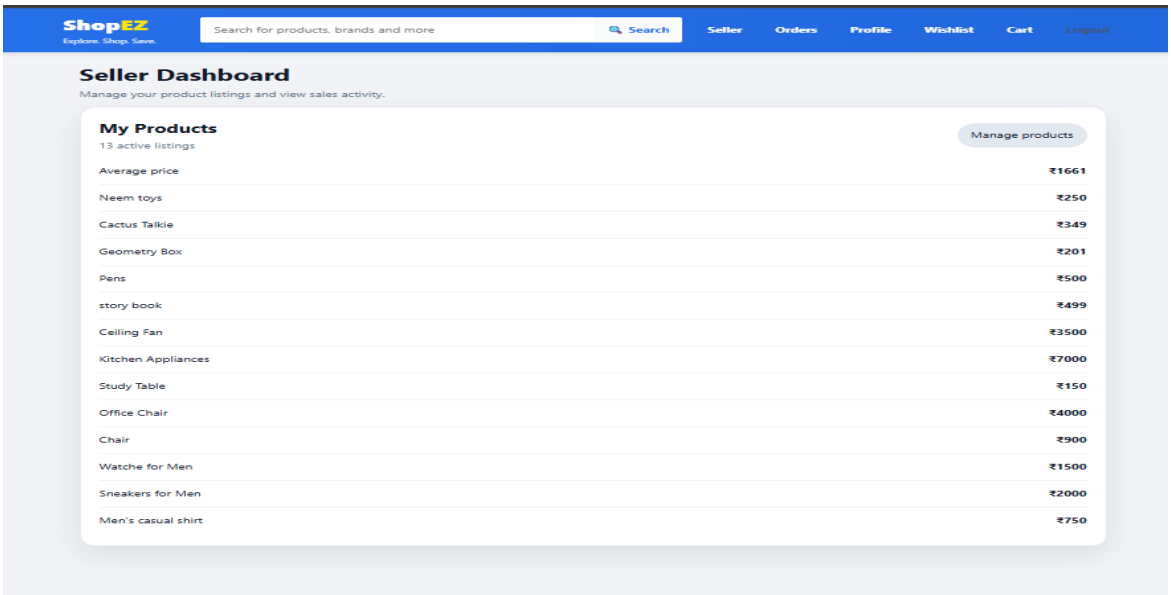
**9. User Interface:**


Shopping Cart
My Cart
1 item
BRAND
Cactus Talkie
₹349
₹401
15% off
Delivery by tomorrow | ₹40 delivery charges
1
Remove
Save for later
Price Details
Price (1 item) ₹349
Discount -₹17
Delivery Charges ₹40
Total Amount ₹372
You will save ₹17 on this order
Place Order


Enter delivery address & payment
Close
Complete your order with a shipping address and preferred payment method.
Full name
sri
Phone
Street address
City
State
Pincode
Country
India
Payment method
Cash On Delivery
Cancel
Confirm & Place Order
Order summary
Price ₹349
Discount -₹17
Delivery ₹40
Total ₹372
You save ₹17 on this order.


My Orders
Order #bb92b1
Payment
pending
Pending
Processing
Shipped
Delivered
Items
Hoorn toys
Qty: 1
Order #163c94
Payment
paid
Items
Smart Watch Pro
Qty: 1
Write Review
Submit Review


15% off
ELECTRONICS
Smart Watch Pro
20 in stock
Track fitness, messages and health metrics.
₹1899
15% OFF
You save ₹285
Color:
Black
Silver
Add to Cart
Buy Now
Available offers
Bank offer 10% instant discount
Free delivery on orders above ₹500
Easy returns and exchange
Customer Reviews
Latest
sri
04/07/2026
good
★ 4
Share your review
Rating
5 stars
Comment
Tell us what you think
Submit review
Rating Breakdown
4.0
1 verified reviews
5★
4★
3★
2★
1★
0%
100%
0%
0%
0%




# Product Management

Add or update products available through your seller/admin account.

## Add new product

Title

Description

Category

Select category

Price

0

Discount (%)

0

Stock

1

Brand

Image URLs

Main image URL

Image URL 2

Second image URL

Image URL 3

Third image URL

Add product

New product

## My Listings

13 products

**Neem toys**
₹250 • Stock: 23
Edit Delete

**Cactus Talkie**
₹349 • Stock: 59
Edit Delete

**Geometry Box**
₹201 • Stock: 30
Edit Delete

**Pens**
₹500 • Stock: 52
Edit Delete

**story book**
₹499 • Stock: 100
Edit Delete

**Ceiling Fan**
₹3500 • Stock: 45
Edit Delete

**Kitchen Appliances**
₹7000 • Stock: 24
Edit Delete

**Study Table**
₹150 • Stock: 50
Edit Delete

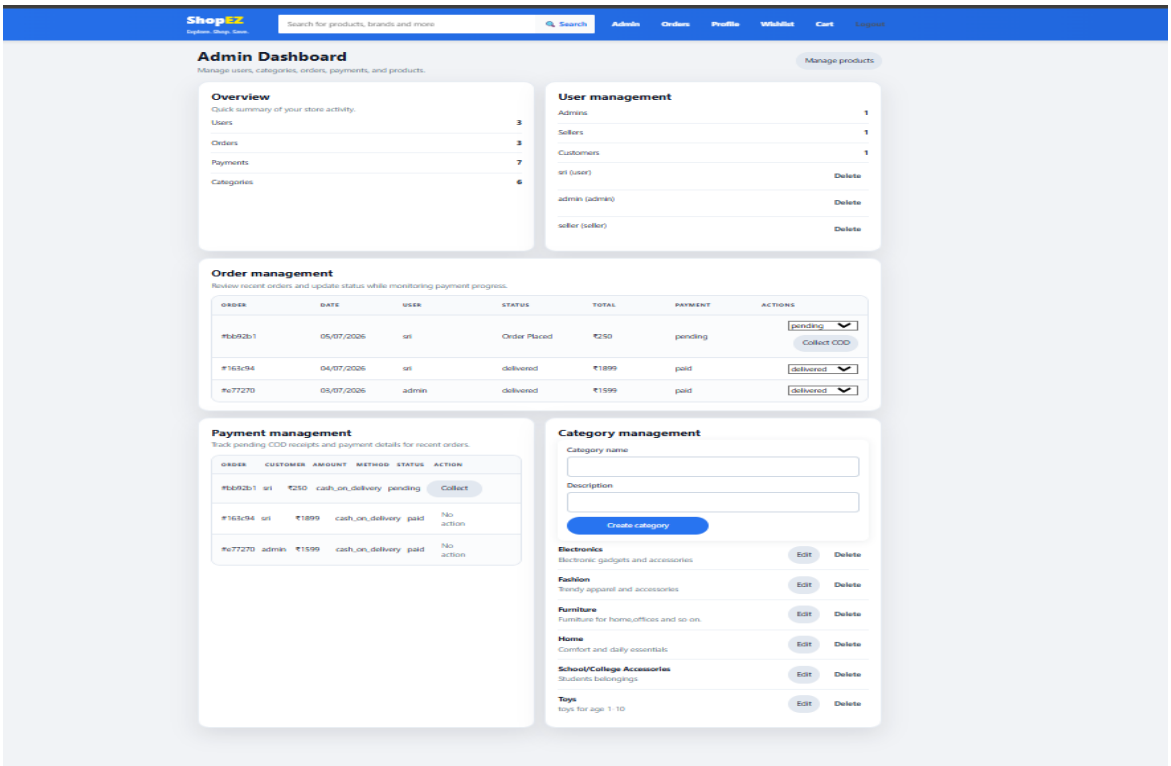
**Office Chair**
₹4000 • Stock: 15
Edit Delete

**Chair**
₹900 • Stock: 12
Edit Delete

**Watche for Men**
₹1500 • Stock: 32
Edit Delete

**Sneakers for Men**
₹2000 • Stock: 25
Edit Delete

**Men's casual shirt**
₹750 • Stock: 11
Edit Delete




## 10. Testing

The project was tested using both manual and API testing methods.

### Functional Testing

- Registration
- Login
- Product CRUD
- Search
- Filters
- Cart
- Wishlist
- Buy Now
- Orders
- Admin Features

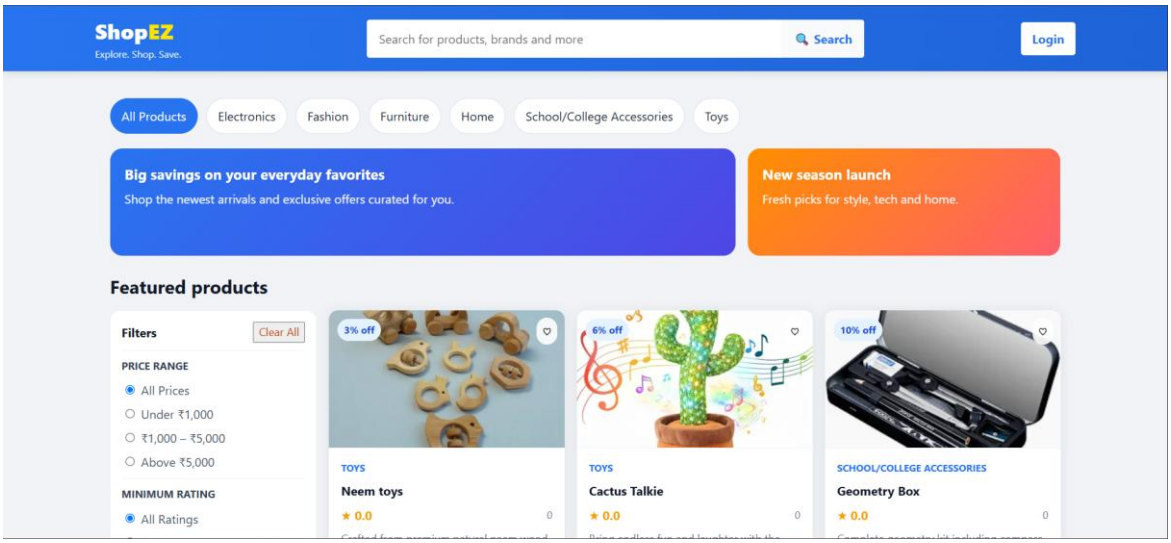
### API Testing

- Postman

### Browser Testing

- Google Chrome
- Microsoft Edge

## 11. Screenshots or Demo :




## 12. Known Issues :

Only Cash on Delivery (COD) payment is currently supported.

Product images are stored using image URLs instead of cloud storage.

Email notifications are not implemented.

## 13. Future Enhancements :

- Razorpay / Stripe Payment Integration
- Product Reviews and Ratings
- Email Verification
- Forgot Password
- Order Tracking
- Coupon & Discount System
- Product Recommendation System
- AI-Based Search
- Multi-language Support
- Cloud Deployment using Vercel and Render
- Sales Analytics Dashboard
- Push Notifications

- Invoice Generation
- Mobile Application Version